Final Lecture: Course Summary, Gaps, & Future Directions

ELG 5218 – Uncertainty Evaluation in Engineering Measurements and Machine Learning

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Outline

1. Part 1: What We Covered
2. Part 2: Uncertainty Quantification Methods
3. Part 2: What We Did *Not* Cover
4. Part 3: Future Directions in UQ
5. Part 4: Course Summary

Uncertainty Decomposition and Implication

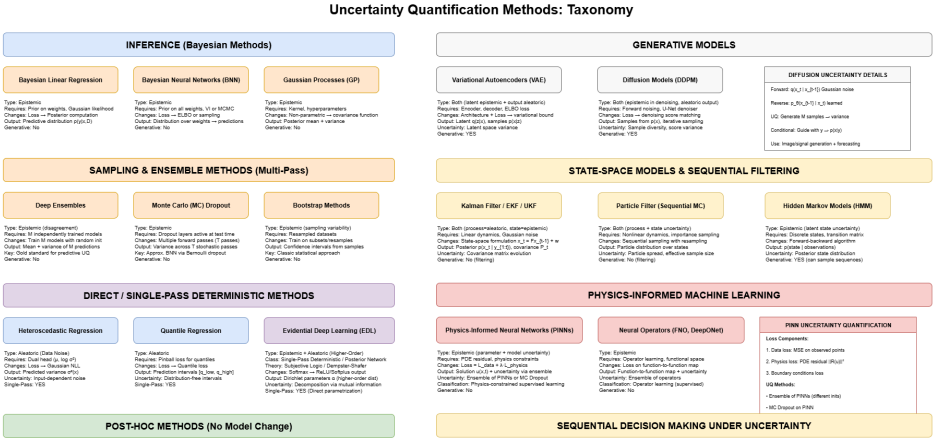
Total Uncertainty = **Aleatoric** + **Epistemic**

Epistemic	Aleatoric	Interpretation	Action
High	Low	Need more data	Collect data / Active learning
Low	High	Task inherently ambiguous	Set safety margins / Wider intervals
Low	Low	High confidence	Proceed with confidence
High	High	Unknown unknowns	Reject or request human review

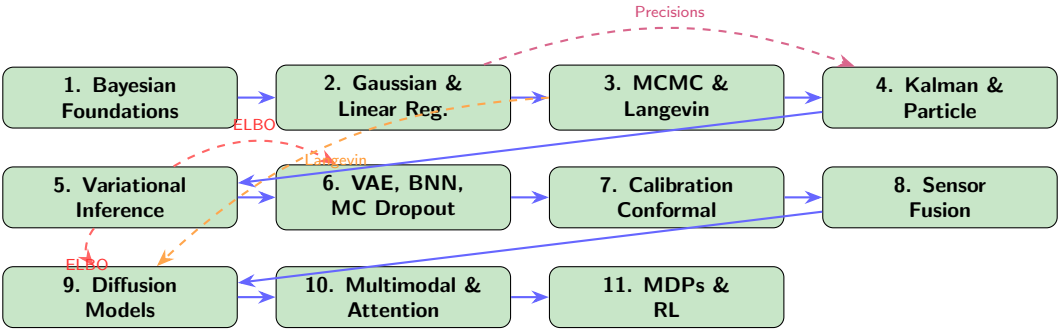
Practical Decision Making:

- **High Epistemic** → *Slow down, collect more data*
- **High Aleatoric** → *Predict well, set wider intervals*
- **Both Low** → *Confidence, proceed*
- **Both High** → *Reject, human intervention*

Taxonomy


Uncertainty Quantification Methods: Taxonomy
INFERENCE (Bayesian Methods)
Bayesian Linear Regression
Type: Epistemic
Requires: Prior on weights, Gaussian likelihood
Changes: Loss → Posterior computation
Output: Predictive distribution p(y|x,D)
Generative: No
Bayesian Neural Networks (BNN)
Type: Epistemic
Requires: Prior on all weights, VI or MCMC
Changes: Loss → ELBO or sampling
Output: Distribution over weights → predictions
Generative: No
Gaussian Processes (GP)
Type: Epistemic
Requires: Kernel, hyperparameters
Changes: Non-parametric → covariance function
Output: Posterior mean + variance
Generative: No
SAMPLING & ENSEMBLE METHODS (Multi-Pass)
Deep Ensembles
Type: Epistemic (disagreement)
Requires: M independently trained models
Changes: Train M models with random init
Output: Mean+ variance of M predictions
Key: Gold standard for predictive UQ
Generative: No
Monte Carlo (MC) Dropout
Type: Epistemic
Requires: Dropout layers active at test time
Changes: Multiple forward passes (T passes)
Output: Variance across T stochastic passes
Key: Approx. BNN via Bernoulli dropout
Generative: No
Bootstrap Methods
Type: Epistemic (sampling variability)
Requires: Resampled datasets
Changes: Train on subsets/resamples
Output: Confidence intervals from samples
Key: Classic statistical approach
Generative: No
DIRECT / SINGLE-PASS DETERMINISTIC METHODS
Heteroscedastic Regression
Type: Aleatoric (Data Noise)
Requires: Dual heads (μ, σ²)
Changes: Loss → Gaussian NLL
Output: Predicted variance σ²(x)
Uncertainty: Input-dependent noise
Single-Pass: YES
Quantile Regression
Type: Aleatoric
Requires: Pinball loss for quantiles
Changes: Loss → Quantile loss
Output: Prediction intervals [q_low, q_high]
Uncertainty: Distribution-free intervals
Single-Pass: YES
Evidential Deep Learning (EDL)
Type: Epistemic + Aleatoric (Higher-Order)
Requires: Dirichlet Prior Network / Posterior Network
Changes: Final layer → Dense(4)+Softplus
Output: Dirichlet parameters α (higher-order dist)
Uncertainty: Decomposition via mutual information
Single-Pass: YES (Direct parametrization)
POST-HOC METHODS (No Model Change)
Conformal Prediction
Type: Neither (Distribution-free coverage)
Requires: Calibration set
Changes: Wrapper around ANY predictor
Output: Prediction sets with guaranteed coverage
Key: Valid regardless of model quality
Generative: No
Temperature Scaling
Type: Neither (Calibration only)
Requires: Validation set
Changes: Scale logits by scalar T
Output: Calibrated probabilities
Key: Simple, effective for confidence
Generative: No
Isotonic Regression
Type: Neither (Calibration only)
Requires: Validation set
Changes: Fit monotonic mapping
Output: Calibrated probabilities
Key: Non-parametric calibration
Generative: No
GENERATIVE MODELS
Variational Autoencoders (VAE)
Type: Both (latent epistemic + output aleatoric)
Requires: Encoder, decoder, ELBO loss
Changes: Architecture + Loss → variational bound
Output: Latent q(z|x), samples p(x|z)
Uncertainty: Latent space variance
Generative: YES
Diffusion Models (DDPM)
Type: Both (epistemic in denoising, aleatoric output)
Requires: Forward noising, U-Net denoiser
Changes: Loss → denoising score matching
Output: Samples from p(x), iterative sampling
Uncertainty: Sample diversity, score variance
Generative: YES
DIFFUSION UNCERTAINTY DETAILS
Forward: q(x_t | x_{t-1}) Gaussian noise
Reverse: p_θ(x_{t-1} | x_t) learned
UQ: Generate M samples → variance
Conditional: Guide with y → p(x|y)
Use: Image/signal generation + forecasting
STATE-SPACE MODELS & SEQUENTIAL FILTERING
Kalman Filter / EKF / UKF
Type: Both (process+aleatoric, state+epistemic)
Requires: Linear dynamics, Gaussian noise
Changes: State-space formulation x_t = Fx_{t-1} + w
Output: Posterior p(x_t | z_{1:t}), covariance P_t
Uncertainty: Covariance matrix evolution
Generative: No (filtering)
Particle Filter (Sequential MC)
Type: Both (process + state uncertainty)
Requires: Nonlinear dynamics, importance sampling
Changes: Sequential sampling with resampling
Output: Particle distribution over states
Uncertainty: Particle spread, effective sample size
Generative: No (filtering)
Hidden Markov Models (HMM)
Type: Epistemic (latent state uncertainty)
Requires: Discrete states, transition matrix
Changes: Forward-backward algorithm
Output: p(state | observations)
Uncertainty: Posterior state distribution
Generative: YES (can sample sequences)
PHYSICS-INFORMED MACHINE LEARNING
Physics-Informed Neural Networks (PINNs)
Type: Epistemic (parameter + model uncertainty)
Requires: PDE residual, boundary conditions
Changes: Loss = L_data + λ L_physics
Output: Solution u(x,t) + uncertainty via ensemble
Uncertainty: Ensemble of PINNs or MC Dropout
Classification: Physics-constrained supervised learning
Generative: No
Neural Operators (FNO, DeepONet)
Type: Epistemic
Requires: Operator learning, functional spaces
Changes: Loss on function-to-function map
Output: Function-to-function map + uncertainty
Uncertainty: Ensemble of operators
Classification: Operator learning (supervised)
Generative: No
PINN UNCERTAINTY QUANTIFICATION
Loss Components:
1. Data loss: MSE on observed points
2. Physics loss: PDE residual ||R(u)||²
3. Boundary conditions loss
UQ Methods:
- Ensemble of PINNs (different inits)
- MC Dropout on PINN
SEQUENTIAL DECISION MAKING UNDER UNCERTAINTY
POMDPs (Partially Observable MDPs)
Type: Epistemic + Aleatoric (state + outcome)
Requires: State space S, actions A, observations O, transition T, observation Z, reward R
Changes: Planning over belief states b_t
Output: Policy π(b_t) and value estimates V(b_t)
Uncertainty: Belief distribution over hidden state
Generative: YES (simulated trajectories)
Bayesian RL / Thompson Sampling
Type: Epistemic (model/parameter)
Requires: Prior over dynamics or value function
Changes: Sampled models/policies for exploration
Output: Policy with exploration driven by uncertainty
Uncertainty: Posterior over model or value estimates
Generative: YES (simulate rollouts under sampled models)

The Course at a Glance: 11 Topics, One Thread



Unifying thread: How to quantify, propagate, and act on uncertainty in ML systems.
Dashed lines: cross-topic connections (ELBO, Langevin/score, precision-weighting).

Core Concepts: The Big Ideas

1. **Aleatoric vs. Epistemic Uncertainty**
   - Aleatoric: irreducible noise (σ^2, Q, R)
   - Epistemic: reducible with data (Σ_N, $P_{t|t}$, ensemble disagreement)
2. **Bayes' Rule is the Engine**
   - Posterior $\propto$ likelihood $\times$ prior
   - "Precisions add" (Gaussian case) unifies Topics 1, 2, 4, 8
3. **The ELBO: A Universal Objective**
   - Appears in VI, VAE, BNN (Bayes by Backprop), and DDPM
   - Reconstruction + KL regularisation trade-off
4. **Score Functions and Langevin Dynamics**
   - From MCMC sampling (Topic 3) to diffusion models (Topic 9)
5. **Calibration Governs Trust**
   - Miscalibrated $\Rightarrow$ wrong Kalman gains, wrong fusion weights, wrong decisions

Methods Summary: What Captures What?

Method	Aleatoric	Epistemic	Single Pass	Scalable	Guarantees
Bayes. Lin. Reg.	✓	✓	✓	Small D	Exact
Gaussian Processes	✓	✓	✓	$O(n^3)$	Exact
MCMC / Langevin	✓	✓	–	Moderate	Asymptotic
Kalman Filter	✓	✓	✓	$O(n_z^3)$	Exact (LG)
Particle Filter	✓	✓	–	Low d	Asymptotic
Mean-Field VI	–	✓	✓	High	Approx.
VAE	✓	partial	✓	High	Approx.
BNN (BbB)	–	✓	–	Moderate	Approx.
MC Dropout	–	✓	–	High	Approx.
Deep Ensemble	partial	✓	–	High	None
Heteroscedastic	✓	–	✓	High	None
Conformal Pred.	–	–	✓	High	**Finite-sample**
Diffusion Models	✓	partial	–	High	Approx.

No single method is perfect: each has trade-offs in expressiveness, scalability, and guarantees.

Cross-Topic Connections: Summary Table

Theme	**Topics Connected**
Precisions add	Gaussian models → Bayes. regression → Kalman → Sensor fusion
ELBO / VI	VI → VAE → BNN → DDPM
Score & Langevin	Langevin MCMC → Score matching → Diffusion reverse
Reparameterization trick	SVI → VAE → BNN
Aleatoric vs. Epistemic	All methods
Calibration → Decisions	Scoring rules → Temp. scaling → Conformal → Fusion → RL
Sequential Bayes	Kalman → Particle → Diffusion → Bellman
Embeddings & Fusion	Contrastive → Cross-attention

UQ Landscape: Covered vs. Not Covered

Covered in depth

- Bayesian inference (conjugate, MCMC, VI)
- Kalman & particle filters
- BNNs, MC Dropout, Deep Ensembles
- Heteroscedastic / MDN networks
- Calibration & proper scoring rules
- Conformal prediction
- VAEs & diffusion models
- Sensor fusion & data association
- Multimodal learning & attention
- MDPs & deep RL (DQN, PPO, SAC)

Not covered (important gaps)

- Gaussian Processes in depth (sparse/scalable GPs)
- Normalizing Flows
- Evidential Deep Learning
- Neural Processes (meta-learning UQ)
- Bayesian optimisation
- Out-of-distribution (OOD) detection
- Uncertainty in LLMs / NLP
- Uncertainty for graphs / GNNs
- Stochastic differential equation models (neural SDEs/ODEs)
- Active learning with UQ
- Causal inference under uncertainty

Important Open Problems: Intuition and Trends

1. **Out-of-Distribution (OOD) Detection**
   - Models are trained on limited data and may fail on unfamiliar inputs
   - OOD detection asks: *"Does this look different from anything I was trained on?"*
   - Ideally, uncertainty should increase for unfamiliar inputs
   - **Current trend:** using learned representations and foundation models to detect OOD more reliably across tasks

2. **Neural SDEs and Continuous-Time Models**
   - Many real systems evolve continuously and are inherently noisy
   - Neural SDEs model both system dynamics and uncertainty over time
   - They produce distributions over future trajectories instead of single predictions
   - **Current trend:** combining SDEs with diffusion models for time-series forecasting and simulation

Important Open Problems (cont.): Intuition and Trends

3. Bayesian Optimization

- Used when experiments are expensive and data is scarce
- A probabilistic model tracks both predictions and uncertainty
- Uncertainty helps decide where to try next
- **Current trend:** scaling Bayesian optimization to high dimensions using neural surrogates and learned acquisition functions

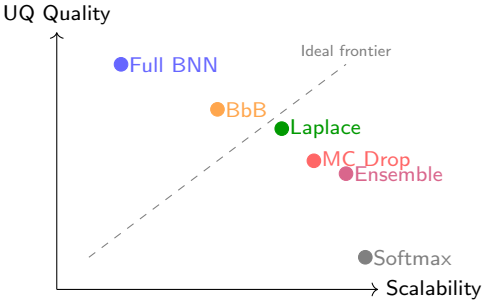
4. Active Learning

- Labels are expensive, but unlabeled data is abundant
- The goal is to label the data points that are most informative
- Uncertainty often indicates where labels would help most
- **Current trend:** combining active learning with deep models and foundation models to reduce human labeling effort

Future Direction 1: Scalable Bayesian Deep Learning

Challenge: Full Bayesian inference over millions of neural network weights is intractable.

- **MC Dropout, Deep Ensembles**
  - Easy to apply and scalable
  - Provide coarse but useful uncertainty estimates
- **Bayes-by-Backprop (BbB)**
  - Fully Bayesian: learns a distribution over every weight
  - Principled, but doubles the number of parameters and is hard to scale
- **Laplace Redux (Daxberger et al., 2021)**
  - Post-hoc Bayesian treatment of the *last layer only*
  - Treats deep networks as deterministic feature extractors + Bayesian linear model
  - Strong uncertainty estimates at a fraction of the cost



Open problems:

- **Subnetwork uncertainty:** Which layers or weights actually matter for predictive uncertainty?
- **Efficient posterior structure:** Can uncertainty be represented using low-rank or structured approximations?
- **Function-space views:** Can we model uncertainty over functions rather than over weights?

Future Direction 2: UQ for Foundation Models and LLMs

Why this matters: LLMs can produce fluent answers that sound convincing even when they are wrong. The main challenge is to estimate *when the model should be trusted*.

Five emerging ideas (grouped by where uncertainty comes from):

1. **Meaning-level uncertainty**
   Semantic entropy: sample several answers, group them by meaning, and measure how spread out the meanings are
2. **Formal guarantees**
   Conformal language generation: wrap the LLM to produce outputs or claims with statistical coverage/correctness guarantees
3. **Agreement across samples**
   Self-consistency: generate multiple reasoning paths and trust answers that repeatedly agree
4. **The model's own self-report**
   Verbalized confidence: ask the model how sure it is, then calibrate that score
5. **External evidence**
   Retrieval-augmented UQ: ground the answer in retrieved documents and assess how reliable the supporting sources are

Connection to this course:

- **Calibration and scoring rules:** do confidence scores match actual correctness?
- **Conformal prediction:** gives finite-sample guarantees for black-box models
- **Bayesian viewpoint:** helps interpret uncertainty over answers, meanings, and evidence

Future Direction 3: UQ for Autonomous Systems

Pipeline: Perception $\rightarrow$ Tracking $\rightarrow$ Fusion $\rightarrow$ Planning $\rightarrow$ Decision

Key challenge: Uncertainty must *propagate* through the entire pipeline.

- **Perception**: Diffusion-based detection (DiffusionDet) with built-in UQ
- **Tracking**: Kalman/particle filters with calibrated detection confidence
- **Fusion**: Multimodal (camera + LIDAR + radar) with learned embeddings and attention
- **Planning**: MDPs/POMDPs with epistemic-aware exploration
- **Decision**: Expected utility with conformal safety guarantees

This course covered every component: Topics 4 (tracking), 7 (calibration), 8 (fusion), 9 (diffusion), 10 (multimodal), 11 (decisions).
Open problem: End-to-end uncertainty propagation from raw sensors to safe decisions.

Future Direction 4: Diffusion Models as Uncertainty Tools

Main idea: Instead of producing one prediction, diffusion models can generate **many plausible solutions**. The spread of these samples gives a rich picture of uncertainty.

Where this is useful:

- **Inverse problems:** MRI, CT, inpainting, compressed sensing
  $\rightarrow$ generate several reconstructions consistent with the measurements
- **Time series:** forecasting and imputation
  $\rightarrow$ generate many future trajectories, then form uncertainty bands
- **Scientific computing:** PDEs with missing or noisy information
  $\rightarrow$ generate multiple possible solution fields
- **Object detection / molecules:**
  $\rightarrow$ sample spread reflects uncertainty in boxes or 3D conformations

Why this is exciting:

- Uncertainty is **sample-based**, not limited to a Gaussian mean/variance form
- Can represent **multiple plausible answers** (multimodal uncertainty)
- Works naturally for complex objects: images, trajectories, fields, 3D structures

Main caution:

- Sampling is computationally expensive
- Diverse samples do *not automatically* mean calibrated uncertainty
- In many applications, diffusion-based posterior sampling is still approximate

Future Direction 5: Conformal Prediction at Scale

Why conformal prediction is growing

- Provides **distribution-free guarantees** that hold for any model
- Guarantees finite-sample coverage, not just asymptotic or average behavior
- Works as a post-hoc wrapper around existing models

Key research frontiers

1. **Conditional coverage**
   How can we give guarantees for specific subgroups or individual inputs, not just on average?
2. **Online and adaptive conformal**
   How can coverage guarantees be maintained as data distributions change over time?
3. **Conformal + diffusion models**
   How can we calibrate generative models so their outputs come with formal guarantees?
4. **Structured outputs**
   How can conformal sets be defined for complex predictions like images, objects, or time series?
5. **Decision-making with conformal sets**
   How should decisions change when predictions are sets rather than point estimates?

Big picture connection

Conformal prediction is the only method in this course that guarantees correct uncertainty calibration at finite sample sizes. Combined with Bayesian models, it offers both interpretability and formal guarantees.

Practical Takeaways for Engineers

1. **Always separate aleatoric from epistemic** – they require different actions (wider intervals vs. more data)
2. **Calibrate your models** – use reliability diagrams, ECE, and temperature scaling before deployment
3. **Consider conformal prediction** as a distribution-free safety net
4. **Ensemble methods are easy** – simple, effective, embarrassingly parallel
5. **For sequential data, use state-space models** – Kalman/particle filters propagate uncertainty naturally
6. **For generative tasks, diffusion models** provide built-in uncertainty via sample diversity
7. **In decision-making under uncertainty**, use expected utility with calibrated probabilities
8. **Check your fusion weights** – miscalibrated sensors propagate errors downstream

Final Exam — Information

Instructions

- **Duration:** 3 hours. **Total:** 100 points + 15 bonus points.
- **Policy:** Closed book. Formula sheet will be provided. Non-programmable calculator allowed.
- Show reasoning clearly; partial credit is awarded.
- No coding questions. Focus: theory, evaluation, interpretation, and small analytical examples.

Question Structure

- **Q1:** Short questions covering the whole course.
- **Q2:** A question from the last topics not covered in the midterms.
- **Q3:** A small example related to assignments.
- **Q4:** An integrative question combining multiple topics.
- **Q5:** Computation, analysis, and interpretation of model results.

Thank You!

Questions?

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Course materials:
https://github.com/Health-Devices/Course-Uncertainty-Machine-Learning-2026